SEV-1

LAB 06: The App Can’t Reach the DB

AWS Day 2 Operations & Troubleshooting · NorthBank

Incident Post-Mortem & Resolution Playbook

NotebookLM

Incoming Alert: Internet Banking is Timing Out

Metadata

Reporter: Marco (App Team)

Severity: SEV-1

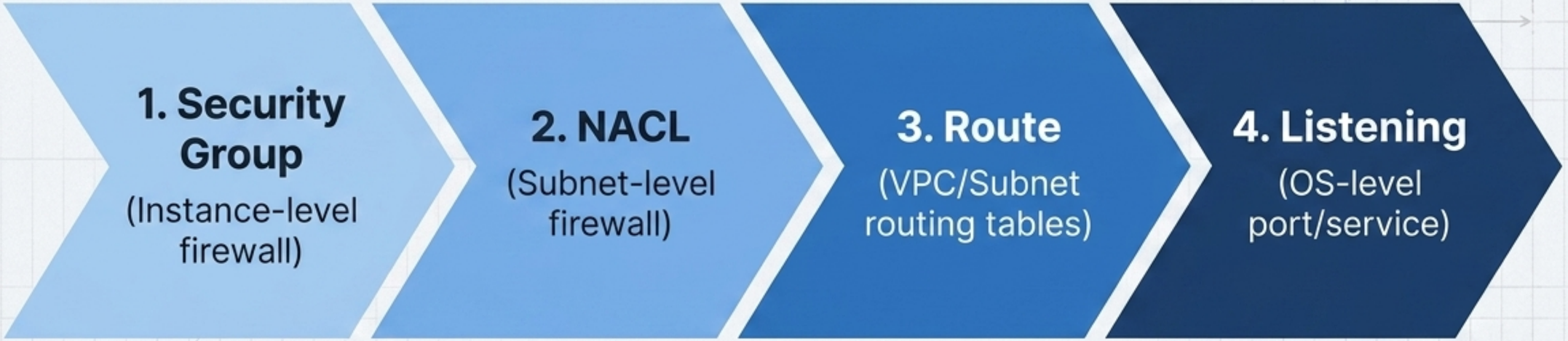
Symptom

“Internet banking is timing out. Targets unhealthy, but the instances look fine.”

Diagnostic Discrepancy: The instances report healthy internally, but external targets are failing.

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Walk the Path in Order, Don't Poke Randomly



Always prove connectivity systematically by tracing the packet's journey layer by layer.

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Know Your Firewalls Before Digging

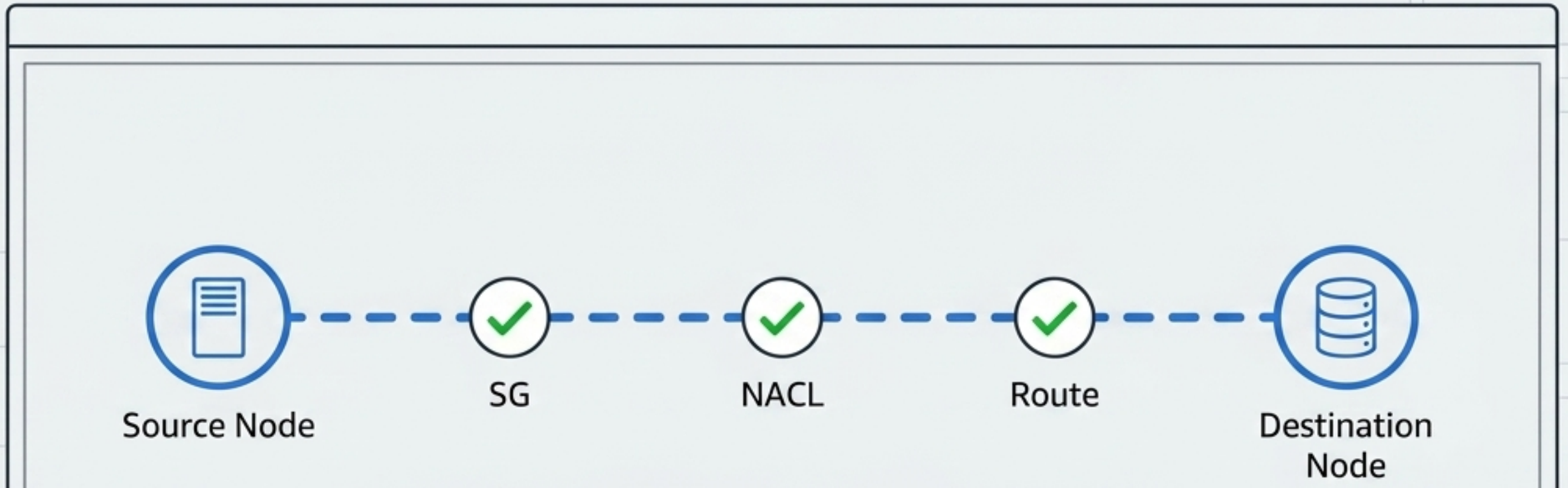
Security Groups	NACLs
Instance-Level	Subnet-Level
Stateful (Return traffic is automatically allowed)	**Stateless** (Requires explicit return rules)

Diagnostic Rule: If an outbound request goes out but fails to return, check statefulness first.

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Prove Connectivity Without Guesswork

Use VPC Reachability Analyzer



Statically analyze the network path to definitively prove where connectivity is breaking across SGs, NACLs, and Routes before making a single configuration change.

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Reproducing the Failure State in the Environment

(The Action)

```
$ export APP_SG, ALB_SG from terraform output
$ ./break.sh
$
```

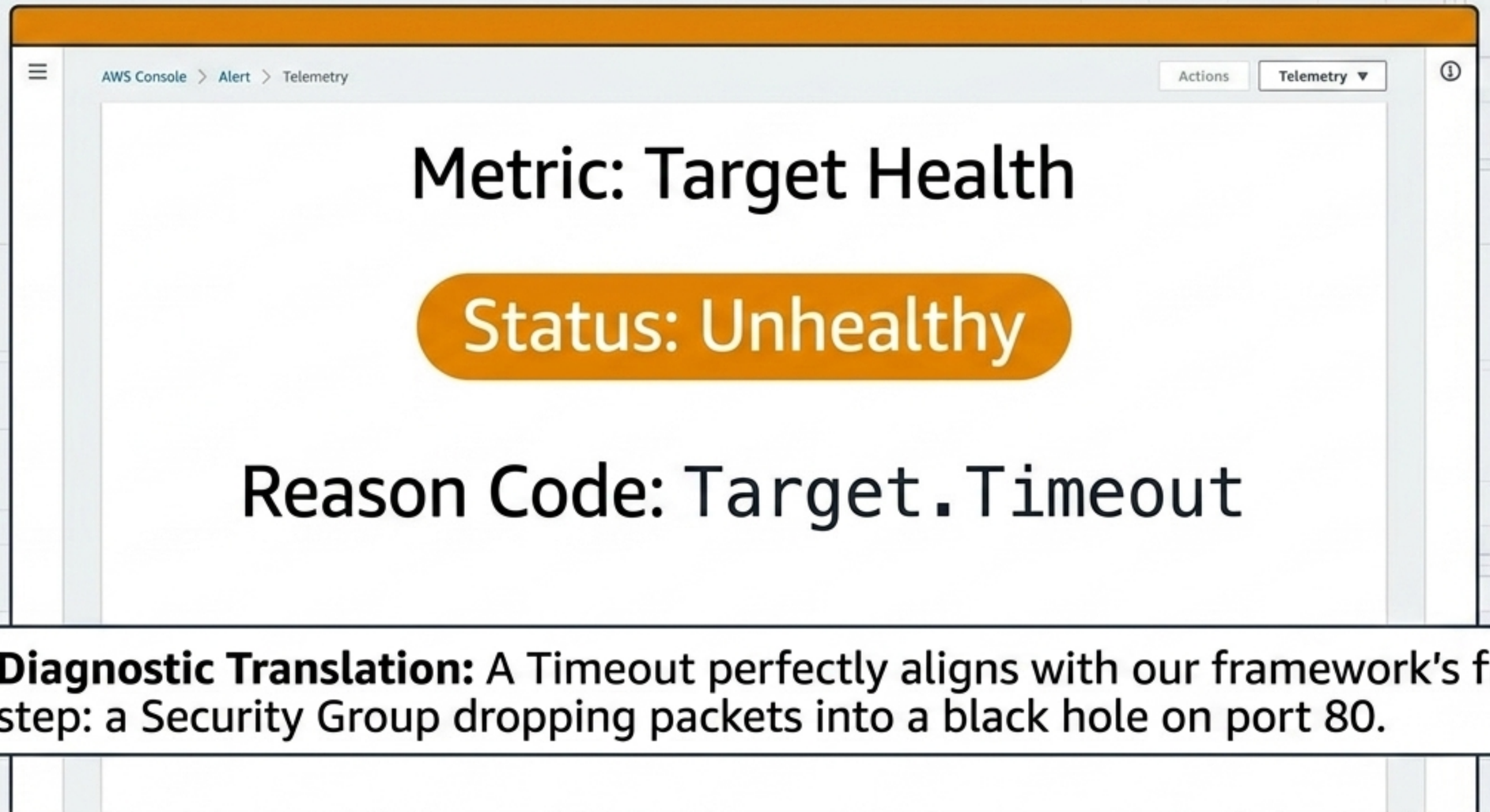
(The Impact)



Action: The break.sh script actively removes the critical ALB → App port-80 rule.

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Tracking the Symptom to the Load Balancer



Diagnostic Translation: A Timeout perfectly aligns with our framework's first step: a Security Group dropping packets into a black hole on port 80.

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The Discovery: Port 80 is Missing from the Security Group

```
$ aws ec2 describe-security-groups
[
   {
      "IpPermissions": [

      ]
   }
]
```



The Finding: The output confirms the port-80 ingress rule is **entirely gone** from APP_SG. The path is **broken** at Step 1.

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The Day-2 Architecture Principle: The Narrowest Fix

The Anti-Pattern



Anti-Pattern: Never use 0.0.0.0/0 just to make it work.

The Standard



Standard: Fix with the narrowest rule—always source a specific Security Group, not an open IP range.

"The narrowest fix today prevents next month's security finding."

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Executing the SRE-Grade Remediation

```
$ aws ec2 authorize-security-group-ingress \
      --group-id $APP_SG \
      --protocol tcp --port 80 \
      --source-group $ALB_SG
```

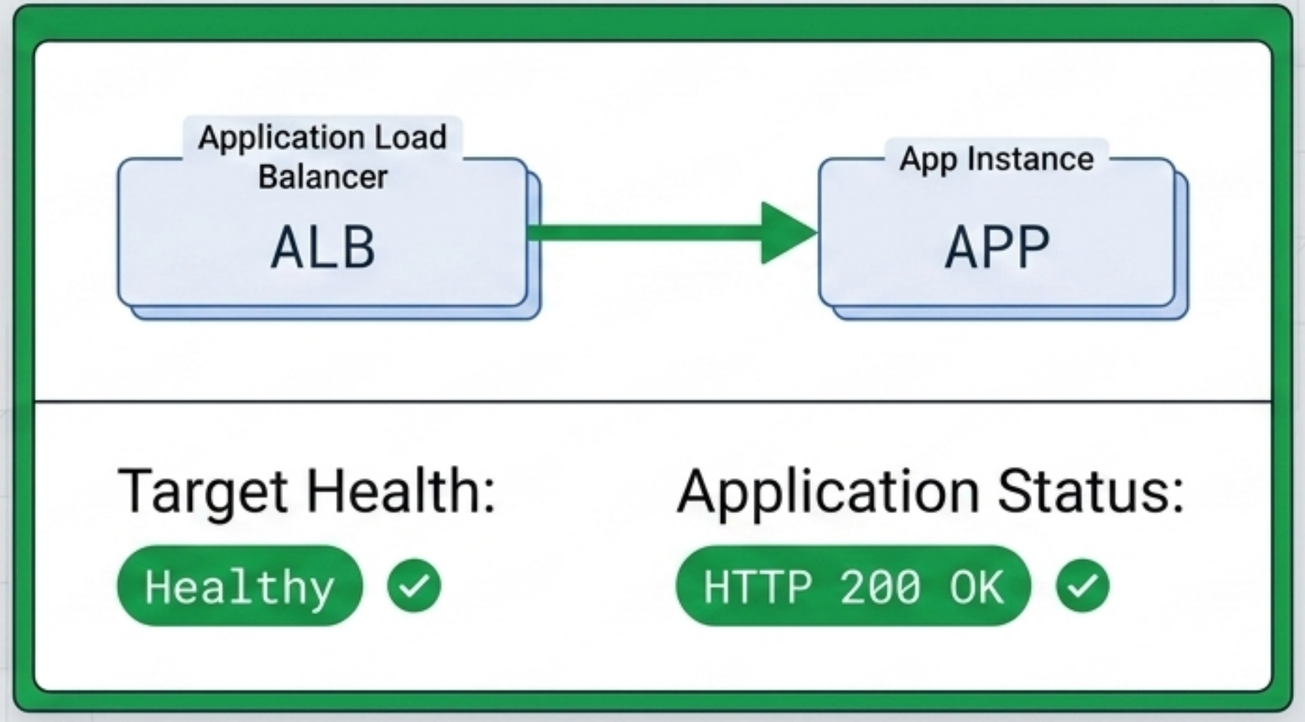
Targeting the broken instance group.

Re-opening the exact listening port.

Applying the narrowest fix (scoping to the ALB SG).

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Validating the Fix and Closing the Ticket



SEV-1 Resolved

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Lab 06 Retrospective & Day-2 Lessons



1. Systematic Triage

Walk the path in order (Security Group → NACL → Route → Listen). Don’t poke randomly.



2. Statefulness

Remember that SGs are stateful, and NACLs are stateless.



3. Least Privilege

The narrowest fix applies. Source security rules to specific SG IDs, never 0.0.0.0/0.

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